FDTD Benchmarks of Luminescent against Lumerical and Tidy3D in Silicon Photonics

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Abstract

We benchmark Luminescent FDTD, a differentiable GPU solver, on silicon photonic devices against a published comparison of Lumerical and Tidy3D [1]: a directional coupler, a 2×2 MMI and a mode converter. On all devices Luminescent exhibits excellent accuracy (within 2% of the three-solver mean), good speed (8000 Mcells/s/gpu), and superior grid convergence which matters because cost grows as $O(N^4)$ in resolution N.

1 Introduction

The finite-difference time-domain (FDTD) method [3, 4, 5] is widely used to simulate silicon photonic components, and GPU solvers have made 3D simulations much faster [6]. Liu and Poon [1] benchmarked Lumerical FDTD [7] and Tidy3D [8] on standard GDSFactory [2] components across resolutions of 6–25 cells per wavelength. We repeat three of their cases with Luminescent [13], using the same layouts and materials, so users can compare the three solvers on equal terms.

2 Setup

All devices are 220 nm silicon waveguides in SiO_2 cladding. Si uses a lossless Lorentzian fit to the Palik model [9] of Ref. [1] ($n = 3.4729$ at 1550 nm); SiO_2 has constant $n = 1.4425$. The fundamental TE mode is launched at the input port, and transmission is read from mode monitors. All boundaries are perfectly matched layers [10]. Luminescent runs on one NVIDIA RTX PRO 6000 Blackwell GPU. Ref. [1] ran Tidy3D on cloud GPUs and Lumerical on a 24-core workstation.

3 Results

3.1 Directional coupler

The device is GDSFactory's `coupler` with a 0.236 µm gap and a 20 µm coupling length. Figure 2(a) shows the cross-port transmission at 1550 nm, and Fig. 2(b) shows the spectrum at 15 cells per wavelength. Luminescent stays between 41% and 44% from 6 to 15 cells per wavelength. Over the same range, Tidy3D gives 45–70% and Lumerical 41–49%. Table 1 lists runtimes.

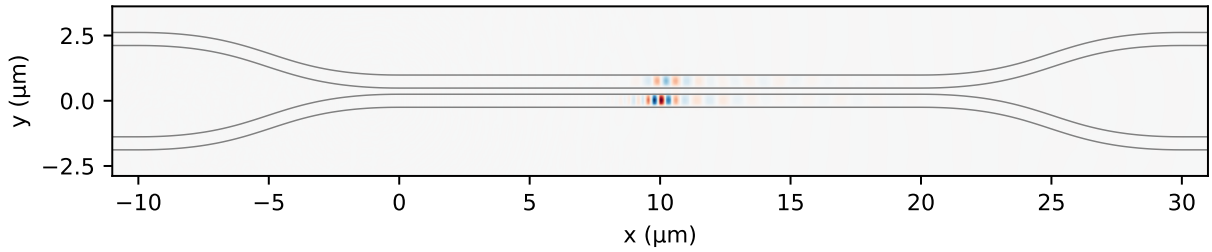


Figure 1: Directional coupler: H_z in the waveguide plane with the pulse mid-device (after 60 optical periods, 15 cells per wavelength).

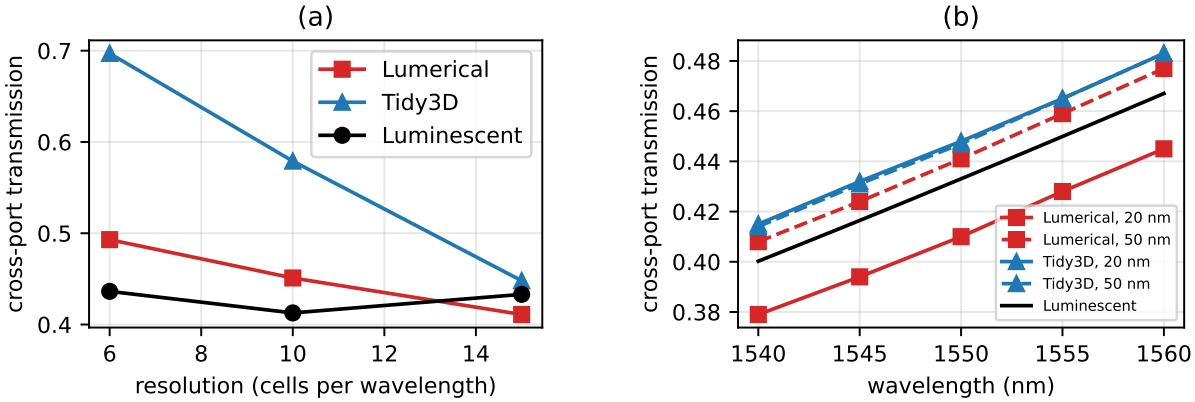


Figure 2: Directional coupler. (a) Cross-port TE0 transmission at 1550 nm against resolution; Lumerical and Tidy3D values digitized from Ref. [1], Fig. 4(a). (b) Cross-port spectrum at 15 cells per wavelength; Lumerical and Tidy3D curves for 20 nm and 50 nm source bandwidths digitized from Ref. [1], Fig. 4(b), within our 1540–1560 nm range.

Table 1: Directional coupler runtime (s). Lumerical and Tidy3D values from Ref. [1], Table 2.

resolution	Luminescent	Tidy3D	Lumerical
6	13	17	148
10	23	35	336
15	133	66	1293

3.2 2×2 MMI

The device is GDSFactory's `mmi2x2_with_sbend`: 500 nm arms tapered to 700 nm before the multimode region, with S-bends at all four ports [11]. Figure 4(a) shows the cross-port transmission at 1550 nm, and Fig. 4(b) shows the spectrum at 15 cells per wavelength. Luminescent agrees with both solvers to within one percentage point at 10 and 15 cells per wavelength. Table 2 lists runtimes.

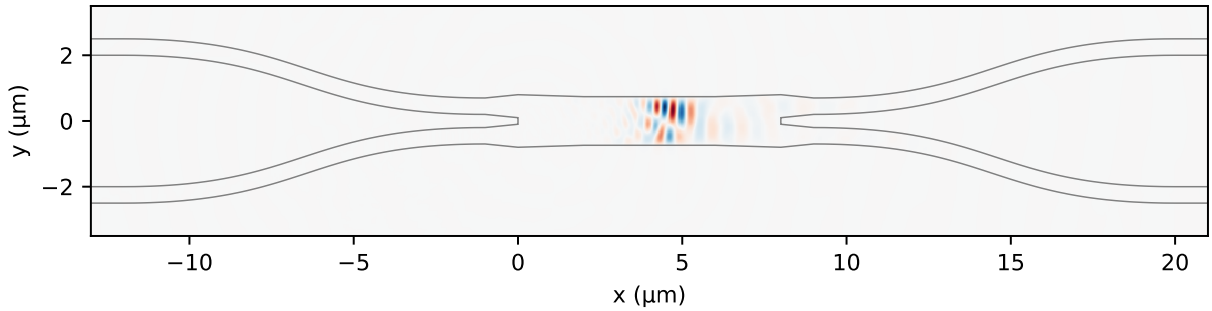


Figure 3: 2×2 MMI: H_z in the waveguide plane with the pulse mid-device (after 50 optical periods, 15 cells per wavelength).

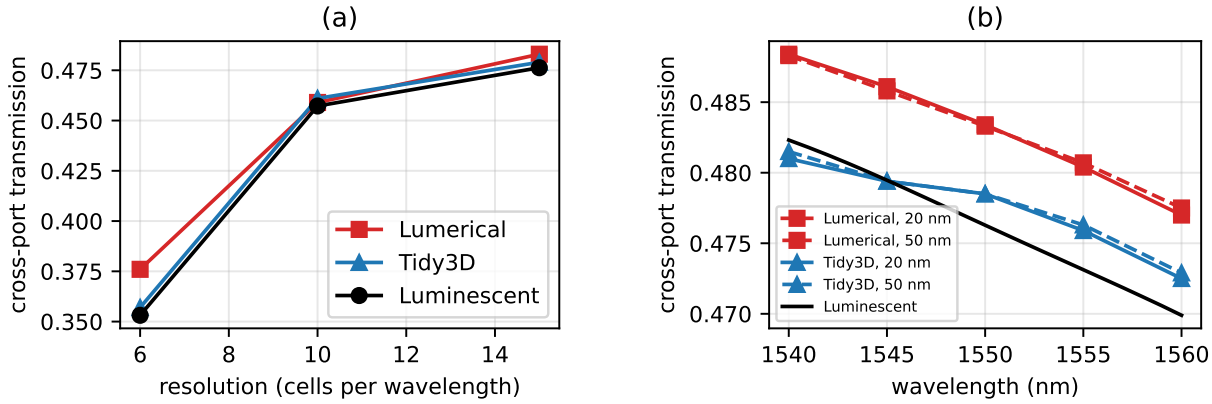


Figure 4: 2×2 MMI. (a) Cross-port TE0 transmission at 1550 nm against resolution; Lumerical and Tidy3D values digitized from Ref. [1], Fig. 12(a). (b) Cross-port spectrum at 15 cells per wavelength; Lumerical and Tidy3D curves for 20 nm and 50 nm source bandwidths digitized from Ref. [1], Fig. 12(b), within our 1540–1560 nm range.

Table 2: 2×2 MMI runtime (s). Lumerical and Tidy3D values from Ref. [1], Table 6.

resolution	Luminescent	Tidy3D	Lumerical
6	12	11	98
10	20	34	259
15	53	45	905

3.3 Mode converter

The device is GDSFactory's `mode_converter` with a 0.15 µm gap and a 20 µm coupling length; its S-bend uses two 45° Euler bends, as in Ref. [1]. TE0 enters the 500 nm arm and couples to TE1 in the 1 µm arm [12]. Figure 6 shows the TE1 transmission at the cross port at 1550 nm, and Table 3 lists runtimes. Luminescent gives 0.50, 0.51 and 0.46 at 6, 10 and 15 cells per wavelength; at 15 it lies between the published values (0.49 Tidy3D, 0.44 Lumerical).

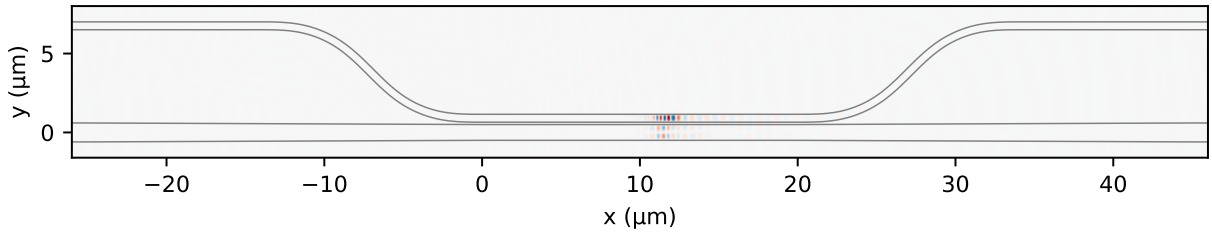


Figure 5: Mode converter: H_z in the waveguide plane with the pulse mid-device (after 80 optical periods, 15 cells per wavelength).

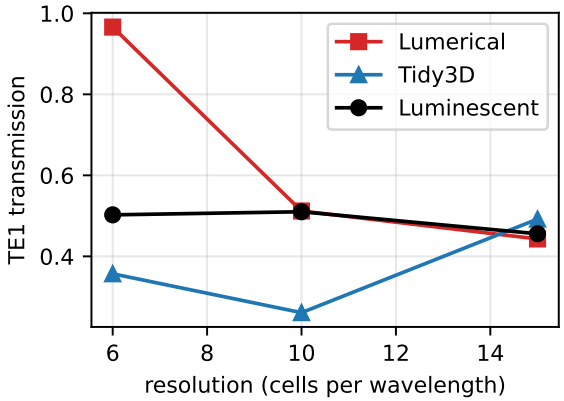


Figure 6: Mode converter: cross-port TE1 transmission at 1550 nm against resolution. Lumerical and Tidy3D values digitized from Ref. [1], Fig. 16(a).

Table 3: Mode converter runtime (s). Lumerical and Tidy3D values from Ref. [1], Table 8.

resolution	Luminescent	Tidy3D	Lumerical
6	26	63	760
10	86	517	3599
15	334	415	14417

4 Conclusion

Table 4: Grid convergence (change of the key result from 10 to 15 cells per wavelength) and agreement (difference from the mean of the three solvers at 15 cells per wavelength), in %. Key result: cross-port transmission (TE1 for the mode converter). Lumerical and Tidy3D values from Ref. [1].

	grid convergence (%)			accuracy agreement (%)		
device	Luminescent	Tidy3D	Lumerical	Luminescent	Tidy3D	Lumerical
directional coupler	4.7	29.2	9.7	0.5	4.0	4.6
2×2 MMI	4.0	3.8	5.0	0.7	0.1	0.7
mode converter	11.9	47.0	15.6	1.7	6.1	4.4

Table 5: Speed (Mcells/s). Luminescent: time-stepping at 15 cells per wavelength. Tidy3D and Lumerical: grid points × iterations / runtime at 25 cells per wavelength, from Ref. [1], Tables 3, 7 and 9.

	Luminescent	Tidy3D	Lumerical
hardware	1 RTX PRO 6000	≤8 A100	24-core CPU
directional coupler	7670	17700	425
2×2 MMI	7880	20700	427
mode converter	8120	36300	441

All three solvers agree within 6% at 15 cells per wavelength. Luminescent has the best grid convergence on the coupler and the mode converter, and is on par with the others on the MMI. Grid convergence matters because cost grows as the fourth power of resolution (cells in 3D times time steps): reaching the same accuracy at 10 instead of 15 cells per wavelength makes a run about five times cheaper. Luminescent runs at 7,700–8,100 Mcells/s on one GPU. Tidy3D's speed is aggregated over up to eight A100 GPUs; for the mode converter, run on eight, it is about 4,500 Mcells/s per GPU. Because the hardware differs, these speeds compare what a typical user gets from each tool, not necessarily the quality of the solvers. Overall Luminescent demonstrates excellent accuracy, good speed, and superior grid convergence on these silicon photonics devices.

References

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